



COMMODITY PRICE FORECASTING • XGBOOST REGRESSOR

Forecasting Commodity Markets with XGBoost

BUSINESS-READY PREDICTIVE ANALYTICS FOR PRICING, PROCUREMENT, AND RISK DECISIONS

5

Separate forecasting tracks for Gold, Silver, Aluminium, Nickel, and Uranium

3

Lag windows combined with rolling means and calendar signals

1

Unified forecasting workflow framed for commercial decision support

• BUSINESS PROBLEM

Why commodity forecasting matters

MARKET VOLATILITY

Commodity prices react quickly to inflation, industrial demand shifts, and macroeconomic uncertainty.

Static forecasting logic often misses turning points when drivers interact in nonlinear ways.

PLANNING PRESSURE

Procurement teams, pricing owners, and investors need forward-looking visibility rather than reactive reporting.

Forecast accuracy directly affects purchase timing, margin protection, and portfolio strategy.

COMMODITY TREND EXAMPLE



Illustrates uneven price movement, sharp reversals, and trend persistence that demand a model capable of nonlinear learning.

BUSINESS OUTCOMES ENABLED

Procurement planning aligned to expected market direction.

Pricing strategy strengthened by forward-looking commodity scenarios.

Investment and exposure decisions supported by predictive signals.

XGBoost pipeline and engineered signal design

By: Liannaka Dadi

DATA

Monthly commodity price history

Gold, Silver, Aluminium, Nickel, Uranium



FEATURE ENGINEERING

Lag 1-3, rolling mean, year, month, quarter



MODELING

Independent XGBoost regressors by commodity

FEATURE STACK

Lag Features

Months 1, 2, and 3 capture short-horizon price memory.

Rolling Means

Smooths noise while preserving directional trend context.

Temporal Features

Year, month, and quarter encode seasonality and macro timing.

Model Design

Per-commodity models isolate market-specific behavior patterns.

ILLUSTRATIVE FEATURE IMPORTANCE



Visualization is presentation-grade and intended to communicate the relative role of recency, smoothing, and temporal context in the model pipeline.

Business impact today, roadmap for scale

VALUE

Nonlinear price behavior captured more effectively

XGBoost strengthens the project's ability to represent abrupt market shifts and interaction effects.

INSIGHT

Historical price dependencies surfaced as leading indicators

Lag structure and rolling context expose how prior movements influence upcoming prices.

USE CASES

Procurement optimization, commodity forecasting, risk monitoring, and investment analysis move from descriptive to predictive decisioning.

SCALABILITY

The architecture supports further model expansion and dashboard integration without changing the business framing.

COMMODITY COVERAGE HEATMAP

	LAG	TREND	SEASON	MODEL FIT
Gold	High	High	Med	Strong
Silver	High	Med	Med	Strong
Aluminium	High	High	Low	Strong
Nickel	High	High	Low	Strong
Uranium	Med	Med	High	Good

PROJECT EVOLUTION ROADMAP

EDA

Market pattern discovery

Feature Engineering

Lag, rolling, temporal signals

Random Forest & XGBoost

Ensemble regression phase

Next

ARIMA, LSTM, live dashboards